

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	JAGADISH.V	Reg. No.:	192521352
Section / Batch:	B.TECH IT	Date:	28/07/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

Q57. A transportation system calculates distances between multiple bus stops to identify the nearest stops using brute-force.

- Explain how all possible pairs are evaluated.
- Derive the time complexity.
- Discuss why the method is inefficient for large inputs.

Code of Conduct

I JAGADISH.V (192521352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.JAGADISH

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

a) Explain how all possible pairs are evaluated

In the brute-force method, each bus stop is compared with every other bus stop exactly once.

Algorithm

1. Start with the first bus stop.
2. Compare its distance with every remaining bus stop.
3. Move to the second bus stop and compare it with the remaining stops.
4. Continue until all unique pairs have been evaluated.
5. Keep track of the smallest distance found.
6. The pair with the minimum distance is reported as the nearest bus stops.

Example

Suppose the bus stops are:

A, B, C, D

The possible pairs are:

- (A, B)
- (A, C)
- (A, D)
- (B, C)
- (B, D)
- (C, D)

For n bus stops, the total number of unique pairs is:

$$\frac{n(n-1)}{2}$$

Each pair is checked once to determine the minimum distance.

b) Derive the Time Complexity

The algorithm uses two nested loops.

for $i = 1$ to $n-1$

for $j = i+1$ to n

compare distance (i, j)

Number of Comparisons

$$\begin{aligned} & (n-1) + (n-2) + (n-3) + \dots + 1 \\ &= \frac{n(n-1)}{2} \end{aligned}$$

Ignoring constants in Big-O notation:

$$T(n) = O(n^2)$$

Time Complexity

- **Best Case:** $O(n^2)$
- **Average Case:** $O(n^2)$
- **Worst Case:** $O(n^2)$

The algorithm always evaluates every possible pair, regardless of the input.

Space Complexity

$$O(1)$$

Only a few variables are needed to store the current minimum distance and the corresponding pair.

c) Discuss why the method is inefficient for large inputs

The brute-force approach becomes inefficient as the number of bus stops increases because the number of comparisons grows quadratically.

Reasons

- Every bus stop is compared with every other bus stop.
- Many comparisons become unnecessary as the dataset grows.
- Execution time increases rapidly for large values of n .
- The algorithm cannot skip comparisons even if a very small distance has already been found.
- It is unsuitable for real-time transportation systems with thousands of bus stops.

Example

Number of Bus Stops (n) Comparisons $\frac{n(n-1)}{2}$

10	45
100	4,950
1,000	499,500
10,000	49,995,000

As **n** increases, the number of comparisons grows dramatically, leading to higher execution time.